

Description

In ancient Egypt, people used the sum of unit fractions (like $1/a$, where a is a positive integer) to represent a real fraction.

Such as: $2/3=1/2+1/6$, but $2/3=1/3+1/3$ is not allowed, because not all the denominators differ from each other.

There are many ways to express a fraction a/b , but which one is the best?

First, the number of terms should be minimized; second, the largest denominator should be minimized.

For example:

$$19/45=1/3 + 1/12 + 1/180$$

$$19/45=1/3 + 1/15 + 1/45$$

$$19/45=1/3 + 1/18 + 1/30$$

$$19/45=1/4 + 1/6 + 1/180$$

$$19/45=1/5 + 1/6 + 1/18$$

The last one is the best solution because 18 is smaller than 180, 45, 30, and 180.

Given a, b ($0 < a < b < 500$), calculate the best solution.

Sample Input

495 499

Sample Output

Case 1: $495/499=1/2+1/5+1/6+1/8+1/3992+1/14970$